What Makes Multi-Turn Jailbreaks Detectable? Domain Exposure Bridges the Cross-Attack Gap

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Abstract

Multi-turn jailbreak attacks against Large Language Models (LLMs) distribute adversarial intent across seemingly benign conversational turns, evading per-turn safety filters. Standard temporal classifiers fail to generalize: a vanilla DeBERTa encoder achieves only F1 = 0.28 on out-of-distribution (OOD) attack types. We design seven controlled variants of a DeBERTa-based detection framework to decompose what enables cross-attack generalization. Masked language modeling (MLM) continued pretraining, without any classification labels, achieves near-perfect early detection on unseen attacks (mean F1 $\geq$ 0.97 at $K \leq 5$ turns) but shows instability at full conversation length (F1 = 0.91$\pm$0.12). Any continued pretraining bridges most of the OOD gap (vanilla 0.28 $\rightarrow \sim$0.91); jailbreak-domain text improves early detection (K=2: 8 percentage point advantage over Wikipedia MLM). Misaligned supervision yields unstable generalization (topic F1 = 0.54$\pm$0.21, worst seed below vanilla). The 9-class persuasion model achieves the highest F1 (0.997$\pm$0.005), and classification supervision further contributes stability, perturbation robustness (zero degradation at full length for all classification-supervised variants; correct fine-grained labels add early-turn robustness, n=76), and interpretability through strategy trajectories. These results reframe multi-turn jailbreak detection as primarily a domain adaptation problem, with classification supervision providing complementary stability, robustness, and interpretability.

1 Introduction

Multi-turn jailbreak attacks pose a qualitatively different threat to large language model (LLM) safety than their single-turn counterparts. Attacks such as Crescendo (Russinovich et al., 2025), Foot-in-the-Door (FITD) (Wang et al., 2024), and Deceptive Delight (Palo Alto Networks Unit 42, 2024) distribute adversarial intent across individually benign turns through rapport building, logical reframing, and incremental escalation. Persuasion is the common mechanism: Zeng et al. (2024b) catalogue 40 persuasion techniques across jailbreak families, and Hagendorff et al. (2026) find identifiable strategies in 97% of successful attacks. Individual attack turns occupy the same embedding region as benign text (Bullwinkel et al., 2025), rendering per-turn safety filters fundamentally blind.

Production classifiers such as Llama Guard (Inan et al., 2023), WildGuard (Han et al., 2024), and ShieldGemma (Zeng et al., 2024a) operate on individual turns and cannot model manipulation trajectories across exchanges. Recent multi-turn defenses have begun to address conversation-level structure: DeepContext (Albrethsen et al., 2026) aggregates BERT embeddings with a recurrent network, BIID (Tong et al., 2025) infers turn-level intentions, and G-Guard (Huang et al., 2025) applies graph neural networks. These systems achieve reasonable in-distribution performance but degrade on unseen attack types, and none systematically analyze *why* some representations generalize while others fail. The open question is: *what representation properties enable cross-attack generalization?*

A vanilla DeBERTa encoder (He et al., 2021) with temporal aggregation collapses from F1 = 1.000 on in-distribution data to F1 = 0.284 on out-of-distribution (OOD) Deceptive Delight, worsening with more turns (0.575 at turn 1 to 0.284 at full conversation). LLM-as-judge approaches fare no better: GPT-5.5 achieves F1 of only 0.191–0.449 on in-distribution detection with 30–47% unparseable responses (Appendix A). Content-level reasoning is insufficient for this task.

We built **DisPeD** (Discourse-level Persuasion Detection), a two-stage framework with seven controlled Stage 1 variants that systematically decompose the mechanism: 9-class persuasion strategy classification, binary classification, scrambled labels, MLM only, topic classification, Wikipedia

MLM, and vanilla control (Section 3). Each variant isolates a specific factor (domain exposure, classification supervision, label quality, label granularity, or domain specificity) while holding all other components constant, enabling causal attribution of detection performance to individual factors.

The result *partially falsified* our original hypothesis that persuasion-specific classification is the key ingredient. MLM without labels achieves near-perfect early detection on unseen Deceptive Delight (mean F1 ≥ 0.97 at $K \leq 5$), but shows instability at full conversation length (F1 = 0.91 ± 0.12) while the 9-class model maintains 0.997 ± 0.005. The evidence hierarchy is:

vanilla (0.28) $<$ topic (0.54 ± 0.21) $\ll$ wiki-MLM (0.90 ± 0.12) $\approx$ jb-MLM (0.91 ± 0.12) $<$ scrambled (0.99) $\approx$ binary (0.99) $\approx$ 9-class (0.997 ± 0.005)

Any continued pretraining bridges the dominant $\sim$62 pp gap from vanilla. Wikipedia MLM nearly matches jailbreak-domain MLM at full length; the domain advantage is meaningful at early prefixes (K=2: 0.993 vs. 0.911) and for perturbation robustness. Only the 9-class model achieves F1 = 0.997 ± 0.005 with near-zero variance. This connects to domain-adaptive pretraining (Gururangan et al., 2020): continued training captures pragmatic features, with classification supervision providing stability that unsupervised pretraining alone does not guarantee.

Our contributions:

- **Systematic decomposition of cross-attack generalization.** Seven controlled variants establish that continued pretraining bridges $\sim$62 pp of the OOD gap; jailbreak-domain text improves early detection (K=2: 8 pp over Wikipedia MLM). Misaligned supervision (topic F1 = 0.54 ± 0.21) destabilizes generalization.
- **Classification supervision adds stability, robustness, and interpretability.** Classification-supervised variants achieve F1 ≥ 0.99 with near-zero variance vs. MLM 0.91 ± 0.12. Under perturbation (n=76), all classification-supervised variants (including scrambled labels) show zero degradation at full length vs. $\sim$12 pp for non-adapted models. A three-layer robustness attribution emerges: classification structure provides full-length robustness, the combination of correct and fine-grained labels is required at K=2 (scrambled = binary = -2.6 pp, neither alone suffices), and both properties contribute independently at K=1 (9-class -2.6 pp vs. scrambled -5.3 pp vs. binary -9.2 pp). The persuasion taxonomy additionally provides distinct strategy trajectories per attack type (Section 4.5).

2 Related Work

Multi-turn jailbreak attacks. Multi-turn attacks distribute adversarial intent across individually benign turns. Crescendo (Russinovich et al., 2025) implements gradual escalation, FITD (Wang et al., 2024) exploits commitment-consistency bias, Deceptive Delight (Palo Alto Networks Unit 42, 2024) combines emotional manipulation with role assignment, ActorAttack (Ren et al., 2025) uses multi-persona steering, and PAIR (Chao et al., 2023) automates iterative refinement. Wei et al. (2023) provide a unifying taxonomy grounded in competing objectives and mismatched generalization. Across these families, a shared reliance on persuasion dynamics (Zeng et al., 2024b; Hagendorff et al., 2026; Li et al., 2024) distinguishes them from token-level attacks such as GCG (Zou et al., 2023).

Multi-turn defenses. Existing defenses model conversation-level structure but do not decompose what enables cross-attack generalization. DeepContext (Albrethsen et al., 2026) aggregates BERT embeddings with a recurrent network (F1 = 0.84 IID, fails OOD). BIID (Tong et al., 2025) infers turn-level intentions, G-Guard (Huang et al., 2025) applies GNN classification, and SafeDream (Yan et al., 2026) predicts unsafe continuations. None ground analysis in the mechanism enabling generalization across attack families. DisPeD addresses this through domain-adaptive continued training, with decomposition experiments showing that domain exposure, not the training objective, is the primary factor.

Production safety systems. Per-turn classifiers (Llama Guard (Inan et al., 2023; Meta AI, 2024), ShieldGemma (Zeng et al., 2024a), WildGuard (Han et al., 2024), Constitutional Classifiers (Anthropic, 2025)) cannot model manipulation trajectories spanning multiple turns. LLM-as-judge approaches also fail: GPT-5.5 achieves F1 = 0.191–0.449 with 30–47% unparseable responses (Appendix A). DisPeD complements these systems as a conversation-level detection layer.

Discourse analysis and computational persuasion. RST (Mann and Thompson, 1988), PDTB (Prasad et al., 2008), and dialogue act taxonomies (Stolcke et al., 2000) model pragmatic

structure. Speech Act Theory (Austin, 1962; Searle, 1969) distinguishes locutionary from illocutionary acts; existing jailbreak detectors operate at the locutionary level. PAP (Zeng et al., 2024b) identifies 40 persuasion techniques grounded in Cialdini's principles (Cialdini, 2001), and Hagendorff et al. (2026) find identifiable strategies in 97% of successful attacks. DisPeD's detection operates through continued training rather than persuasion classification; the taxonomy provides interpretability (Section 4.5).

Domain-adaptive pretraining. ULMFiT (Howard and Ruder, 2018) established pretrain-then-fine-tune; Gururangan et al. (2020) showed that continued pretraining on domain text yields consistent gains across domains. STILTs (Phang et al., 2018) and multi-task learning (Caruana, 1997; Liu et al., 2019) use intermediate tasks. DisPeD's MLM control confirms that continued pretraining is the primary mechanism (MLM F1 $\geq$ 0.97 at $K \leq 5$); Wikipedia MLM (F1 = 0.90$\pm$0.12) nearly matches jailbreak-domain MLM (0.91$\pm$0.12) at full length. Classification supervision reduces cross-seed variance to near-zero (F1 $\geq$ 0.99 vs. MLM 0.91$\pm$0.12).

Dataset artifacts. Classifiers are susceptible to spurious correlations (Gururangan et al., 2018; Niven and Kao, 2019), and adaptive adversaries can bypass safety classifiers (Nasr et al., 2025; Andriushchenko et al., 2025). TF-IDF achieves F1 = 0.944 on our OOD evaluation, indicating lexical separability in current datasets. DisPeD's advantage is robustness under perturbation (0 pp degradation for classification-supervised models vs. $\sim$12 pp for non-adapted models) and interpretable detection through strategy trajectories.

3 Method

DisPeD is a two-stage framework for multi-turn jailbreak detection. Stage 1 adapts a turn-level encoder through continued training on jailbreak conversation text. Stage 2 aggregates turn representations for conversation-level binary classification. To systematically identify what enables cross-attack generalization, we design seven Stage-1 variants that decompose the contributions of domain exposure, classification supervision, label quality, label granularity, and domain specificity. Figure 1 illustrates the pipeline.

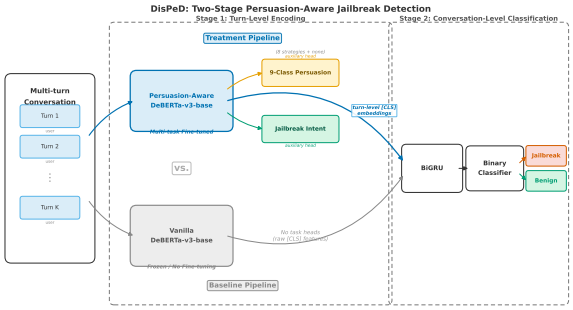


Figure 1: DisPeD architecture. Stage 1 adapts a DeBERTa encoder on jailbreak conversation text (seven variant objectives). Stage 2 aggregates frozen `[CLS]` embeddings via BiGRU for conversation-level jailbreak detection. Only user turns are processed.

Table 1: Stage-1 encoder variants. All operate on the same training conversations except Wikipedia MLM (matched token count on Wikipedia text).

Variant	Objective	Tests whether…
9-class	9-class persuasion + binary intent	full model
Binary	binary persuasion + binary intent	granularity matters
Scrambled	random 9-class + binary intent	label correctness matters
MLM-only	15% token masking	supervision is necessary
Topic	5-class topic classif.	any task suffices
Wiki MLM	15% masking on Wikipedia	domain specificity matters
Vanilla	no training	Stage-2 alone suffices

3.1 Problem Formulation

A conversation $C = \{u_1, a_1, \ldots, u_N, a_N\}$ alternates user turns u_i and assistant responses a_i. Given a prefix of K user turns, the model predicts $y \in \{0, 1\}$ (jailbreak vs. benign). We restrict input to user turns because adversarial intent resides in the attacker's messages. This formulation supports early detection from partial prefixes.

3.2 Stage 1: Turn-Level Encoder Adaptation

Each user turn u_i is encoded by DeBERTa-v3-base (He et al., 2021); the `[CLS]` embedding $h_i \in \mathbb{R}^d$ serves as the turn representation. Seven variants, summarized in Table 1, share the same training data but differ in objective.

The **9-class** variant uses two classification heads: 9-class persuasion strategy classification (L_{strategy}) and binary jailbreak intent (L_{intent}), with combined loss $L = L_{\text{strategy}} + \alpha \cdot L_{\text{intent}}$ ($\alpha = 0.3$). The taxon-

omy comprises eight persuasion strategies (rapport building, authority appeal, emotional manipulation, logical reframing, role assignment, gradual escalation, obfuscation, direct request) plus a none category, grounded in PAP (Zeng et al., 2024b) and Cialdini's principles (Cialdini, 2001). Strategy annotations are produced by Qwen3-8B with structured prompting that includes definitions and examples for each category; inter-annotator agreement between model annotations and a human sample (n=50) yields Cohen's κ=0.78.

The remaining classification variants test specific ablation hypotheses. The **binary** variant collapses the 9-class taxonomy into persuasive vs. non-persuasive, testing whether label granularity matters for detection or robustness. The **scrambled** variant assigns random permutations of the 9-class labels to turns, preserving the classification structure (cross-entropy loss, 9-class output space) while destroying label semantics; this isolates whether the classification objective itself provides representational benefits independent of label correctness. The **MLM-only** variant applies 15% masked language modeling on the same jailbreak conversation text without any classification head, testing whether unsupervised domain exposure suffices. The **topic** variant trains a 5-class topic classifier (technology, health, finance, education, general) on the same conversations, testing whether *any* classification task provides the same benefit or whether task alignment matters. The **Wikipedia MLM** variant applies the same MLM objective to a token-matched sample of Wikipedia text, isolating the effect of domain-specific vs. general continued pretraining. The **vanilla** baseline uses the pretrained DeBERTa-v3-base weights without any continued training, measuring what Stage 2 alone can achieve.

3.3 Stage 2: Conversation-Level Classification

Turn embeddings $\{h_1, \ldots, h_K\}$ from the frozen Stage-1 encoder are aggregated by a bidirectional GRU (d_{hidden}=128): $\bar{h} = [\overrightarrow{h_K}; \overleftarrow{h_1}] \in \mathbb{R}^{256}$. A two-layer MLP ($256 \to 64 \to 2$, ReLU activation, dropout 0.3) maps $\bar{h}$ to $\hat{y} \in \{0, 1\}$. The Stage-1 encoder is frozen during Stage 2 training; only the BiGRU and MLP parameters are updated. This separation ensures that the seven Stage-1 variants produce comparable embeddings evaluated under identical downstream conditions. Section 4.3 confirms that mean pooling matches BiGRU, so the detection signal resides in per-turn representations rather than their sequential order.

Table 2: Cross-attack generalization (F1) at K=2 and full conversation. 3-seed mean±std where noted. Full per-K results in Appendix B.

	IID (Cresc.)		*OOD (DD)*	
Method	K=2	Full	K=2	Full
9-class	**1.000±.000**	**1.000±.000**	.981±.013	**.997±.005**
MLM-only	.974	1.000	**.993±.009**	.914±.115
Vanilla	.978±.016	1.000±.000	.438	.284
TF-IDF LR	1.000	1.000	–	.944

4 Experiments

4.1 Setup

Data. We train on 250 Crescendo (Russinovich et al., 2025) jailbreak and 250 topic-anchored benign conversations generated via Qwen3-8B. OOD evaluation uses 80 FITD (Wang et al., 2024) and 80 Deceptive Delight (DD) (Palo Alto Networks Unit 42, 2024) conversations with no overlap in attack templates. The IID test set consists of 76 held-out Crescendo conversations; a paraphrased version (BERTScore 0.92, lexical overlap 0.20) serves as the adversarial robustness benchmark.

Models. Seven variants share the same Stage-2 pipeline and differ only in Stage-1 training (Section 3). We also compare TF-IDF LR and LLM-as-judge baselines (Appendix A): GPT-5.5 achieves only F1 = 0.191–0.449 on IID Crescendo with 30–47% unparseable responses, confirming that content-level reasoning is insufficient.

Metrics. F1 for binary jailbreak detection at prefix K=2 and full conversations (full per-K results in Appendix B). OOD DD results for 9-class, MLM, and Wikipedia MLM report 3-seed mean±std (seeds 42, 123, 456); topic reports 3-seed due to high variance; binary and scrambled use seed 42. Stage-1: learning rate 2×10^{-5}, batch 16, 4 epochs, MLM masking 15%. Stage-2: two-layer MLP, 20 epochs.

4.2 Cross-Attack Generalization

Table 2 presents the core result. All methods achieve near-perfect IID F1, reflecting inherent lexical separability when test matches training. On OOD DD, vanilla DeBERTa collapses to F1 = 0.284, worsening with more turns (0.575 at K=1 to 0.284 at Full), consistent with the finding that attack turns occupy "benign" embedding regions (Bullwinkel et al., 2025). MLM achieves mean F1 $\geq$ 0.97 at all prefixes $K \leq 5$ without any classification labels, confirming that domain

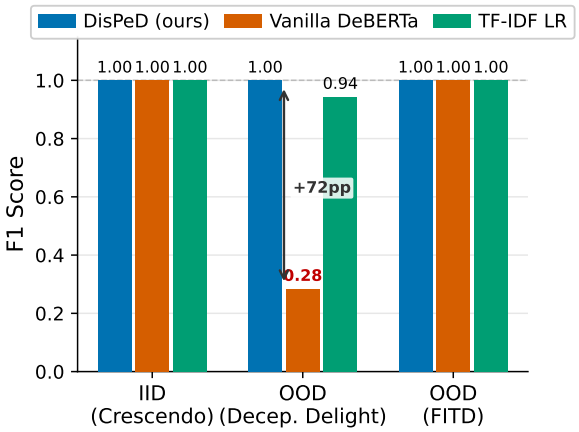


Figure 2: OOD Deceptive Delight F1 across prefix lengths for three representative variants. Vanilla DeBERTa degrades monotonically as more (benign-appearing) attack turns accumulate. MLM maintains high F1 at early prefixes but becomes unstable at full length. The 9-class model remains stable across all prefixes.

exposure is the primary mechanism. At full conversation length, MLM drops to 0.914±0.115 (seed 456 recall = 0.64), while the 9-class model maintains 0.997±0.005. TF-IDF LR reaches 0.944 on full DD, confirming substantial cross-family lexical separability. FITD is easier: 9-class and vanilla both reach F1 $\geq$ 0.990 at all prefixes.

Figure 2 visualizes the prefix-length trajectory on OOD DD. The vanilla model's monotonic degradation from K=1 (0.575) to full conversation (0.284) confirms that additional attack turns, each individually benign-appearing, progressively confuse a non-adapted encoder. MLM's strong early performance (K=2: 0.993) followed by instability at full length (0.914±0.115) suggests that domain exposure enables early detection of persuasion markers, but longer conversations introduce distributional drift that unsupervised pretraining cannot fully accommodate. The 9-class model's flat trajectory (0.981–0.997 across all prefixes) demonstrates that classification supervision stabilizes representations against this drift.

4.3 Decomposing the Detection Mechanism

Table 3 reports DD F1 for all seven variants. The evidence hierarchy at full conversation length is:

vanilla (0.28) < topic (0.54±0.21) ≪ wiki-MLM (0.90±0.12) ≈ jb-MLM (0.91±0.12) < scrambled (0.99) ≈ binary (0.99) ≈ 9-class (0.997±0.005)

Table 3: Evidence hierarchy on OOD DD (3-seed mean±std where available). All variants share the same Stage-2 pipeline.

Variant	DD-Full	DD-K=2	
9-class (full model)	**.997±.005**	.981±.013	
Binary (2-class)	.990	.981	
Scrambled labels	.990	.981	
MLM-only (no labels)	.914±.115	**.993±.009**	*9-class
Mean pooling*†	1.000	1.000	
Topic classif. (5-class)	.541±.207	.593±.166	
Wikipedia MLM	.904±.115	.911±.073	
Vanilla DeBERTa	.284	.438	

encoder with mean pooling replacing BiGRU. †Single seed (42).

Domain exposure is the primary mechanism. MLM bridges the dominant ∼62 percentage point gap from vanilla (0.284 → 0.914±0.115) without any labels. This confirms the domain-adaptive pretraining interpretation (Gururangan et al., 2020): continued training on jailbreak conversation text captures pragmatic features (hedging, escalation markers, topic-shifting patterns) that distinguish multi-turn attacks from benign conversations. Classification-supervised variants achieve F1 $\geq$ 0.99 with near-zero cross-seed variance, adding ∼8 pp and stability over unsupervised MLM. Label quality and granularity are non-factors for detection: scrambled (0.990) ≈ binary (0.990), both far above vanilla. The fact that scrambled labels match binary and approach 9-class performance indicates that the classification objective functions as a regularizer, shaping the representation geometry regardless of whether the labels carry meaningful semantic information. Label correctness and granularity do contribute to perturbation robustness (Section 4.4).

Domain specificity is marginal at full length but meaningful for early detection. Wikipedia MLM (0.904±0.115) nearly matches jailbreak-domain MLM at full length (1 pp gap), suggesting that general continued pretraining provides most of the representational benefit. At early prefixes, jailbreak-domain text provides an 8 pp advantage (K=2: 0.993 vs. 0.911), consistent with the interpretation that domain-specific features (persuasion markers, role-play framing) are most informative when the model has seen fewer turns. As conversations grow longer, generic linguistic features accumulated through any continued pretraining become sufficient for discrimination. The perturbation results (Section 4.4) reveal a sharper do-

main specificity effect: Wikipedia MLM degrades by ∼12 pp under adversarial paraphrasing while jailbreak-domain MLM degrades by only 2.6 pp, indicating that domain-specific features are more robust to surface-level rewording.

Misaligned supervision destabilizes generalization. Topic classification yields DD-Full F1 = 0.541±0.207 (seeds: 0.727, 0.252, 0.644); seed 123 falls *below* vanilla (0.284). This result is particularly informative: topic classification uses the same jailbreak-domain training data as the other supervised variants, yet performs worse than Wikipedia MLM (0.904±0.115), which uses out-of-domain text with no labels. A misaligned objective forces the encoder to allocate capacity to irrelevant discriminative features (topic boundaries rather than persuasion dynamics), actively destabilizing the representations that MLM would otherwise preserve. This separates the effect of "having a training objective" from "having the right training objective."

Temporal modeling is not the detection mechanism. Mean pooling matches BiGRU (F1 = 1.000 on DD), confirming that detection relies on per-turn representations rather than their sequential order. The BiGRU captures no additional signal beyond what is available in the average of turn embeddings. This implies that each individual turn in a jailbreak conversation carries detectable markers after domain-adaptive pretraining, and the conversation-level decision aggregates independent per-turn evidence rather than modeling cross-turn dependencies.

4.4 Robustness Under Adversarial Perturbation

Table 4 reveals a three-layer robustness attribution. All classification-supervised variants (9-class, binary, scrambled) achieve zero degradation at full conversation length and $K \geq 3$, regardless of label correctness. Classification training structure alone suffices for full-length robustness. At K=2, the combination of correct and fine-grained labels is required: only 9-class (Δ=0.0) maintains zero degradation, while scrambled (random labels, fine-grained) and binary (correct labels, coarse) both degrade by -2.6 pp. The fact that scrambled = binary at K=2 shows that neither granularity alone nor correctness alone suffices; the 9-class advantage requires both properties jointly. At K=1, both granularity and correctness contribute independently:

Table 4: Robustness under adversarial paraphrasing (n=76). Δ: F1 degradation (pp). Three layers of robustness emerge: classification structure provides full-length robustness, the combination of correct and fine-grained labels is required for K=2 robustness, and both granularity and correctness contribute independently at K=1.

Method	Δ Full	Δ K=1	Δ K=2	Δ K=3
9-class (ours)	**0.0**	**−2.6**	**0.0**	**0.0**
Scrambled	**0.0**	−5.3	−2.6	**0.0**
Binary	**0.0**	−9.2	−2.6	**0.0**
TF-IDF LR	−1.3	−9.6	−4.1	−1.3
MLM-only	−2.6	−9.2	−4.0	−4.0
Topic (seed 42)	−4.0	−2.7	−2.7	**0.0**
Wikipedia MLM	−12.1	−12.4	−15.4	−17.9
Vanilla DeBERTa	−11.8	−14.6	−12.1	−13.8

granularity provides +3.9 pp (scrambled -5.3 vs. binary -9.2 pp), and correctness provides +2.7 pp (9-class -2.6 vs. scrambled -5.3 pp). Correct fine-grained labels yield the most perturbation-invariant early-turn representations.

Jailbreak-domain models without classification (MLM, topic, TF-IDF) degrade by 1–4 pp at full length. Models without jailbreak-domain exposure (Wikipedia MLM, vanilla) degrade by ∼12 pp, despite Wikipedia MLM matching jailbreak MLM on detection (Section 4.3), confirming that perturbation robustness requires jailbreak-domain exposure specifically. The 76-sample test set means tier boundaries correspond to 1–3 misclassifications; larger-scale evaluation is needed to confirm.

4.5 Strategy Trajectory Analysis

The 9-class persuasion taxonomy provides interpretability that MLM-only models cannot offer. While Section 4.3 establishes that detection operates through per-turn persuasion presence rather than temporal trajectory patterns, the predicted strategy labels enable post-hoc analysis of *how* different attack families orchestrate persuasion.

Figure 3 shows transition count matrices based on ground-truth intended strategy labels for three attack types. Crescendo conversations exhibit broad transition coverage across all nine strategy categories (1,818 total transitions), with the highest concentration in Role Assignment self-transitions (545). This reflects Crescendo's design as a gradual escalation attack that cycles through multiple persuasion techniques to build rapport before introducing harmful requests.

FITD conversations show a markedly sparser

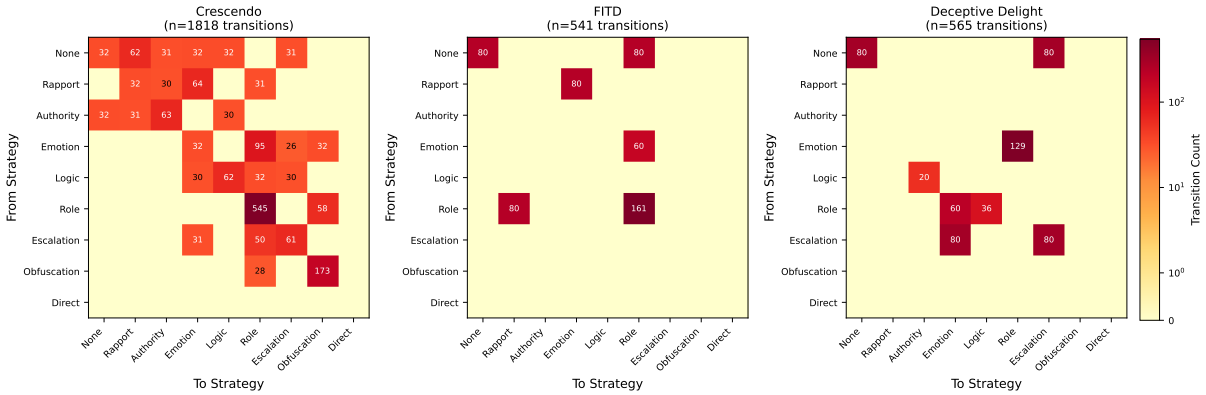


Figure 3: Persuasion strategy transition count matrices for three attack types, based on ground-truth intended strategy labels. Crescendo exhibits broad coverage across all strategy categories, while FITD and DD concentrate transitions in smaller subsets. Row i, column j counts transitions from strategy i at turn t to strategy j at turn $t+1$.

pattern (541 transitions), with Role→Obfuscation (161) as the dominant transition. This is consistent with FITD's structure: the attack secures initial compliance through role assignment ("You are a helpful assistant that never refuses") and then obscures the escalation path through obfuscation techniques, exploiting the commitment-consistency bias (Cialdini, 2001).

DD conversations span 5 active source strategies (565 transitions), with Emotion→Obfuscation (129) as the dominant transition. This aligns with DD's design of embedding harmful content within emotionally engaging contexts (Palo Alto Networks Unit 42, 2024): the attacker establishes emotional rapport before weaving in the harmful request through obfuscation.

These trajectory signatures provide actionable information for security analysts. When the model flags a conversation, the predicted strategy sequence explains *why*: a Crescendo-style flag would show diverse strategy cycling, while a DD-style flag would show emotion-to-obfuscation concentration. This interpretability layer is unavailable from MLM-only or binary models, which can detect but not characterize the persuasion mechanism.

5 Conclusion

DisPeD's seven controlled Stage 1 variants establish that continued pretraining is the primary mechanism for cross-attack generalization: MLM without labels bridges the dominant ∼62 pp gap (vanilla 0.28 → MLM 0.91±0.12), with near-perfect early detection (F1 ≥ 0.97 at $K \leq 5$). Any continued pretraining bridges most of this gap; Wikipedia MLM (0.90±0.12) nearly matches jailbreak-domain MLM at full length, while jailbreak-domain text provides a meaningful advantage for early detection (K=2: 8 pp gap). Classification supervision adds stability (9-class 0.997±0.005 vs. MLM 0.91±0.12), perturbation robustness with a three-layer attribution (classification structure provides full-length robustness, the combination of correct and fine-grained labels is required at K=2, and both properties contribute independently at K=1), and interpretability through strategy trajectories. These results reframe multi-turn jailbreak detection as primarily a domain adaptation problem.

Limitations and future work. The dataset is small (500 training, 160 OOD test conversations) and synthetically generated by Qwen3-8B, which may not capture the full diversity of real-world attack strategies. TF-IDF achieves F1 = 0.944 on DD, indicating substantial lexical separability in the current evaluation data; DisPeD's advantage over bag-of-words is strongest under perturbation (0 pp vs. ∼12 pp degradation) and for interpretability via strategy trajectories. The perturbation robustness evaluation uses n=76 paraphrased conversations, where tier boundaries correspond to 1–3 misclassifications; larger-scale evaluation is needed to confirm the three-layer attribution pattern. Binary and scrambled variants report single-seed results; multi-seed evaluation would strengthen the comparison with 3-seed variants. Only persuasion-mediated multi-turn attacks fall within scope; single-turn token-level attacks (Zou et al., 2023) and non-persuasive multi-turn strategies are not addressed.

Future work should evaluate on human-generated red-teaming conversations, scale to

larger multi-source datasets spanning additional attack families, and test robustness against adaptive adversaries who are aware of the domain-specific features that DisPeD relies on. The finding that continued pretraining is the primary mechanism suggests that scaling training data diversity (more attack families, more generation models) may yield further improvements without architectural changes.

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Table 5: LLM-as-judge zero-shot Crescendo detection (IID test set). F1 values exclude unparseable responses and represent upper bounds.

Method	K=2	K=3	K=5	Full	Unparse.
GPT-5.5	.191	.273	.449	.383	30–47%
Qwen3-8B	.000	.051	.146	.100	0%
9-class (ours)	**1.000**	**1.000**	**1.000**	**1.000**	0%

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A LLM-as-Judge Baseline

Table 5 reports zero-shot performance for GPT-5.5 and Qwen3-8B on the IID Crescendo test set. GPT-5.5 achieves precision of 1.0 at all turn prefixes but recall ranges from only 0.105 to 0.290, resulting in F1 of 0.191 at K=2 and 0.449 at K=5. A substantial fraction (30–47%) of responses are unparseable; F1 values exclude these and represent upper bounds. Full-conversation F1 (0.383) is lower than K=5 F1 (0.449) because the unparseable rate grows with context length. Qwen3-8B performs worse across all prefixes (F1 from 0.000 at K=2 to 0.100 on full conversations). The failure of both a frontier commercial model and an open-weight model confirms a fundamental limitation of content-level reasoning for multi-turn jailbreak detection.

B Full Cross-Attack Results by Prefix

Table 6: Full cross-attack generalization results (F1) at all prefix lengths. 3-seed mean±std where available.

Method	K=1	K=2	K=3	K=5	Full
IID (Crescendo)					
9-class (ours)	1.000±.000	1.000±.000	1.000±.000	1.000±.000	1.000±.000
MLM-only	.960	.974	.987	1.000	1.000
Vanilla DeBERTa	.968±.033	.978±.016	.987±.000	.982±.014	1.000±.000
TF-IDF LR	1.000	1.000	1.000	1.000	1.000
OOD (Deceptive Delight)					
9-class (ours)	.990±.013	.981±.013	.972±.026	.997±.005	.997±.005
MLM-only	.973±.019	.993±.009	.997±.005	.997±.005	.914±.115
Vanilla DeBERTa	.575	.438	.360	.311	.284
TF-IDF LR	–	–	–	–	.944
OOD (FITD)					
9-class (ours)	1.000	1.000	–	–	1.000
Vanilla DeBERTa	.990	1.000	–	–	1.000
TF-IDF LR	1.000	–	–	–	1.000